



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : ME G641
Course Title : Theory of Elasticity & Plasticity
Instructor-in-Charge : J S Rathore

1. Course Description: The course has wide-ranging application in Engineering structures, Machine tool structures, metal forming, and cold working. The studies deal with the determination of stresses and displacements in common structural forms beams, shafts, shells etc. subjected to external forces. The course covers basic theory of elasticity, 2-D and 3-D problems of elasticity, Theories of plastic flow, Flow of ideally plastic and strain hardening metals, and theory of metal forming processes.

2. Course Level Objectives: The course is for higher degree students to prepare them in applying the theory on some problems of practical and design interest. The students will be able to appreciate underlying meaning of material and structural behavior under complex loading conditions. Students are also encouraged to do assignments using the ANSYS/ABAQUS package to solve problems of elasticity and plasticity to compare the analytical solutions.

CO1: Understand the concept of stress and strain in situations where large deformations are involved.

CO2: Appreciate the governing differential equations that describe a general boundary value problem in elasticity.

CO3: Solve the general elasticity boundary value problems with various boundary conditions using methods of complex variable techniques

CO4: Appreciate the concept of yielding and yield criteria. Yield surface and normality rule, associated flow rule

3. Text Books:

T1 Timoshenko S. P. & Goodier J. N., *Theory of Elasticity*, 3rd Edition, Mc-Graw Hill, 1982.

T2 J. Chakrabarty, *Theory of Plasticity*, Mc-Graw Hill Book Company, Singapore, 1987.

4. Reference Books:

R1 M. Filonenko-Borodich, *Theory of Elasticity*, Foreign Language Publishing House, Moscow.

R2 PDS Varma, *Theory of Elasticity*, Vikas Publishing House Private Limited, New Delhi, 1997.

5. Course Plan:

Module	Lecture session	Reference Chap./Sec. # (Book)	Learning Objectives
Introduction to Theory of Elasticity & Plasticity	L1-4: Plane strain and Plane stress of theory of elasticity, Methods of analysis	Ch 1-2 T1	Transformation of stress components. Principle stresses and Principle planes.



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Module	Lecture session	Reference Chap./Sec. # (Book)	Learning Objectives
2-D elasticity problems in Rectangular coordinates	L5-15: Bending of beams	Ch 3 T1	Plane elasticity problems, Biharmonic equation Airy's stress function Solution in rectangular coordinates
2-D elasticity problems in Polar Coordinates	L16-21: General Equation, Pure bending, Rotating Disk, Edge dislocation, Effect of circular hole	Ch 4 T1	Airy's stress function Solution in polar coordinates
3-D elasticity Problems	L22-25: Principal stresses, Stress invariants, Homogeneous deformation, strain at a point, Plastic theory of beding: Stresses beyond elastic limit	Ch 7 T1	Equilibrium equations Strain Compatibility equations Constitutive equations: Elasticity Invariants of Stress tensor
Torsion problems of elasticity	L 26-32: Torsion in straight bars of non-circular cross sections, Hollow sections, Plastic theory of torsion: Stresses beyond elastic limit	Ch 9 T1	Membrane analogy, Narrow rectangular cross section, General solution, torsion of rolled profiles, hollow shafts, thin tubes, screw dislocations
Physics of Plasticity, Steady state problems in plane strain	L33-40: Constitutive relations, Problems contained plastic deformation, True stress-strain curve, strain rate effects, idealization, Yield criteria, plastic stress-strain relations, Symmetrical extrusion, Unsymmetrical extrusion, multi-hole extrusion, Sheet drawing, Cold rolling, mechanics of hot rolling	Ch 2-5 of T2, R3 Ch 7 and 8 R3	Yield surface, Normality rule, Yield criteria Deformation and Flow theories of Plasticity Solution of Plasticity Problems

6. Evaluation Scheme:

Evaluation Component	Duration (Min)	Weightage (%)	Date & Time	Remarks
Mid Sem	90 min	25	As per timetable	CB
Seminar	50 min	20	Weekly	
Project	--	20		
Compre. Exam	180 min	35	As per timetable	CB/OB



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7. **Consultation Hour:** Saturday 11AM to 12 Noon
8. **Notices:** All notices will be communicated via email and put up on Nalanda portal.
9. **Make-up Policy:** Make-up will be granted only to genuine cases. For cases related to illness, proper documentary evidence is essential. Prior permission is necessary.

Instructor-in-charge
ME G641